Supporting Information for "Incomplete river-temperature years are more useful than completeness rules assume"

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Contents

Texts S1–S7, Figure S1 and Tables S1–S22. All tables are generated from the result files by `scripts/463_tr2_evidence.py`. Tables S1–S17 correspond to the tables of the first submission, recomputed where the corrected definitions apply; Tables S18–S22 are new.

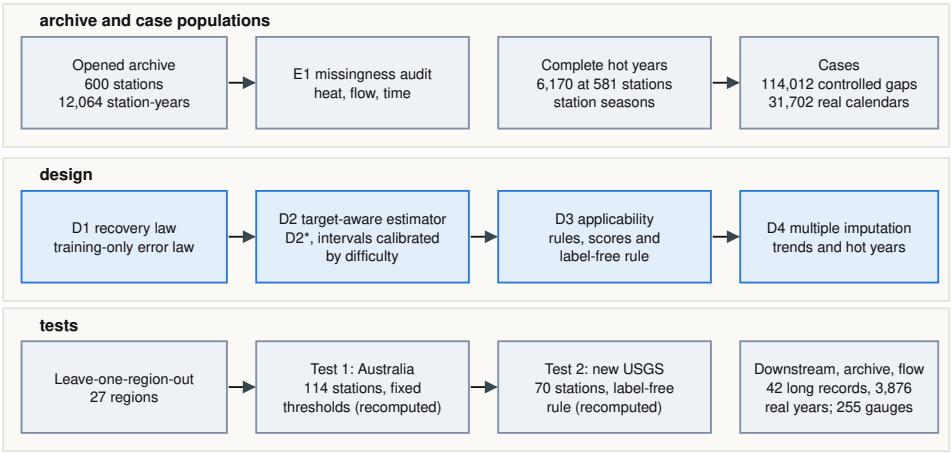


Figure S1. Study design. The archive and case populations feed four design components (D1–D4). They are tested by leave-one-region-out transfer, two preregistered tests on data not used in development (Australia with fixed thresholds; new USGS stations with the label-free rule), both recomputed after their first evaluation (Text S6), a downstream trend experiment, an application to real incomplete years and a low-flow demonstration.

S1 Archive, forcing, regions and hot years

The archive combines daily water temperature from the U.S. Geological Survey, the Bavarian, Baden-Württemberg, Czech, Slovenian, Swiss, Norwegian, Finnish and Dutch agencies, the French Hub'Eau service and the UK Environment Agency, assembled and quality-controlled for an earlier study. A station enters the case populations when it has at least four complete years. Air temperature is the NASA POWER daily 2-m temperature at the station coordinates. For 132 USGS stations added after the original forcing was built, the same product was retrieved from the coordinates in the USGS site service; 11 stations without coordinates have no air layer. Daily discharge (parameter 00060, statistic 00003) was retrieved from the USGS daily-value service; 287 of 336 USGS stations returned discharge. Independence groups join networks that share any station record. Regions are the European agencies and, for USGS stations, the two-digit hydrologic unit from the site service. Transfer analyses use the 14 regions with at least five independence groups.

S1.0.0.1 Hot years and station seasons. Records of stations south of the equator are shifted by 184 days before any year is formed, so that each analysis year runs from July to June; northern records are not shifted. For each case, the smoothed day-of-year climatology of the training years gives the station's five hottest consecutive months (its warm season, used by the warm-season features, the warm-season rules and the recovery-law moments) and its four hottest months (the degree-day target). The climatological peak window is the 30-day window with the highest training climatology; its centre anchors the peak-season rules (below). None of these quantities uses the target year. Table S18 compares the corrected and the original definitions: in the source, masks, true H30, plug-in values and information layers are identical case by case, 163 of 581 stations have a warm season other than May–September, 40 have degree-day months other than June–September, and 11.8% of the cases have a different warm-season observed fraction. In Australia the share of station-years whose H30 window touches the year boundary falls from 31% to 1.1%.

S2 Missingness audit

The station-year sample contains fully operational years from 1982 onwards, excluding the first and last calendar year of each record, with summer (June–August)

air temperature available. The summer anomaly is standardized within station. The conditional logit uses the station as stratum, so all between-station differences in maintenance and climate are removed; the mixed model adds random intercepts for station and for region × year, which absorbs regional shocks such as an agency-wide outage. For the daily analysis, a day is at risk when the station is active, the month is May–September and the previous day was observed; a gap onset is an at-risk day that is missing. Strata of station × year × month compare onset days with non-onset days of the same station-month, which controls for seasonality, station and year by design. The 7-day air anomaly is the departure from a 31-day smoothed day-of-year climatology averaged over the preceding week and standardized by its warm-season standard deviation at the station. Low flow means daily discharge below the 20th percentile of the station's record for that calendar month. Gap duration was regressed on the onset anomaly with station and month effects. Hot summers are the top 10% of years by the region's mean standardized summer anomaly, for regions with at least ten years. The implied bias of a complete-years-only H30 climatology uses each station's regression of water H30 on air H30 in complete years and the difference in mean air H30 between complete years and all years. USGS daily values carry no cause code for missing days (all 68,180 missing days examined in an earlier audit), so gaps could not be split by cause. The audit uses Northern Hemisphere stations only and is unaffected by the corrected definitions. Table S1 lists all models.

S3 Case populations, estimators, interval calibration and learning

S3.0.0.1 Masks. The controlled population places one contiguous 30-, 60- or 90-day gap whose overlap with the training-period climatological 30-day peak window is zero, between 15% and 85%, or complete, twice per cell. The deployment population draws, without replacement, six real incomplete station-years of the same agency (fully operational years with at least one missing day) and moves each of their gaps by a uniform shift of up to 15 days (gaps touching the start or end of the year are not moved). Calendars leaving fewer than 30 observed days are skipped. The agency weights equal the agency's share of real incomplete station-years divided by its share of calendar cases. The state-dependent populations reweight calendar cases by the exact likelihood ratio between onset probabilities proportional to $\exp(k\beta z_t)$ and the uniform shift (main text); β is the case-crossover log odds ratio of

the 7-day air anomaly (1.07 per standard deviation) and its upper 95% bound gives a further stress level.

S3.0.0.2 Plug-in and process estimators. Neighbour regressions use the neighbours observed on at least 90% of the hidden days and two annual harmonics; air regressions use air temperature at lags 0, 1, 2, 3, 5, 7 and 14 days and one harmonic; both are ridge regressions with unit penalty fitted on all non-target days. Air2stream-3 is calibrated by bounded least squares on the longest air-temperature segment outside the target year and integrated over each gap from the preceding observation.

S3.0.0.3 Conditional simulation. The residual history is the out-of-fold residual of the chosen reconstruction on each training year. Raw residual blocks resample 14-day blocks. The air-state law draws a slow 60-day residual path from a training year chosen with weights that compare its residuals on visible days and its air anomaly on the hidden days with those of the target year, and adds seasonally matched 14-day blocks of the fast residual. Each case has 128 draws.

S3.0.0.4 Learning. P2, P5a, P5b and the direct failure classifiers are LightGBM models (300 trees, learning rate 0.05, 31 leaves, at least 50 cases per leaf, 80% row and column subsampling) on decision-time features only: gap geometry, the overlap with the climatological peak, the information layer, donor number and correlation, the recovery-law inputs, the air anomaly over the gap and the warm season, the visible water anomaly next to the gaps, the disagreement among reconstructions, out-of-fold reconstruction diagnostics, and the other reconstructions and the recorded-day value relative to the plug-in. P5b and the classifiers add the quantiles of both simulations relative to the plug-in. Every model is fitted 27 times, leaving out one region, and predicts only the held-out region. Quantile crossing is removed by sorting. For each of the two tests, the same models were refitted on all source cases and frozen.

S3.0.0.5 Interval calibration. For each quantile level τ of D2* and each decision-time group g (share of the climatological peak window hidden: zero, or one of four quarters; information layer: neighbour, air, other, with groups below 300 cases merged into the air layer), the offset is the conformal quantile of the residuals

$y - q_\tau$ of the calibration cases: the $\lceil (n+1)\tau \rceil$-th order statistic for $\tau \geq 0.5$ and the $\lfloor (n+1)\tau \rfloor$-th for $\tau < 0.5$ (Romano et al., 2019). Offsets for a region are estimated on the other regions; the frozen models use offsets estimated on all source cases. Calibrated quantiles are sorted. Table S19 gives the coverage and width of the 90% interval in each group before and after calibration. The preset criterion (85–95% in every group) was not met because the groups with a visible peak over-cover: a large share of their errors is exactly zero, and no quantile of a distribution with a point mass can be placed to cover exactly 90%. The intervals are therefore reported as approximate, with coverage by group.

S4 Recovery law, rules and tolerance sensitivity

S4.0.0.1 Derivation. Let p_t be the reconstruction and $a_t = y_t - s_t$ the anomaly. For a conditional-mean reconstruction, $r_t = p_t - y_t$ is uncorrelated with p_s, so $\mathrm{Cov}(r_t, a_s) = -\mathrm{Cov}(r_t, r_s) = -\sigma_r^2 \phi_r^{|t-s|}$ for an AR(1) residual. The hottest window $\mathcal{S}$ is selected on $\bar{a}_{\mathcal{S}}$, whose variance is $\sigma_a^2 v(30, \phi_a)$. Linear prediction of the hidden residual mean from $\bar{a}_{\mathcal{S}}$ gives the slope κ_m of the main text, and the conditional variance follows from the same covariances. The daily-scale version (v1) replaces κ_m by $1 - R^2 = \sigma_r^2/\sigma_a^2$ and the conditional variance by $\sigma_r^2 R^2 v(m, \phi_r)$; it overstates the shrinkage whenever the residual is less persistent than the anomaly. The peak-window version (v2p) takes the mean and standard deviation of the out-of-fold residuals inside the training years' peak windows instead of the warm-season moments and the shrinkage term. The replay (v3) applies the current mask to each training year, completes the hidden days with that year's out-of-fold residuals and takes the empirical distribution of the resulting H30 errors, smoothed with a Gaussian kernel; it is an empirical comparator, not a law. Tables S3 and S4 compare the versions at case and cell level.

S4.0.0.2 Hard cells. In the five cells with the highest observed failure rate among cells with at least 200 cases, the law predicted 34% against 39% observed (v2p: 39%). The standardized errors $(e - \hat{\mu})/\hat{\sigma}$ exceeded two in absolute value in 11.5% of these cases, the excess kurtosis of the errors was 13 and their variance was 23% larger than the law's. The ratio of the residual standard deviation inside the peak windows to its warm-season value was 0.89, so the shortfall comes from heavy tails rather than from larger residuals in hot periods.

S4.0.0.3 Rules. Rules are applied to the case's missing-day mask in its hot year. The USGS indicator rule requires at least 311 observed days and at least 24 per month; its strict form requires 26 per month and no 7-day window with more than one missing day. The WMO 11/5 rule requires every month to have fewer than 11 missing days and no run of five or more. ETCCDI requires at most 15 missing days in the year and at most three per month; GSIM at least 350 observed days. The peak-season rules evaluate July–September, August, July–August and 15 July–15 September as written for Northern Hemisphere streams whose peak is near 30 July (day 211). We move these windows by the difference between the station's climatological peak centre and day 211, which keeps their position relative to the peak; Table 2 of the main text also reports them as written. The NRFA-type rule scores a year by the position of its best fully observed 30-day mean in the training distribution of H30. Table 2 of the main text adds, for each fixed rule, the failure rate of the plug-in among the rejected years and, for rules with a free threshold, the result when the threshold is chosen on the other regions.

S4.0.0.4 Tolerance and risk. Table S16 repeats the main comparison at 0.25 and 1.0 °C. Retention at 2.5% and 10% risk is in Table S7.

S5 Transfer, the two preregistered tests and post hoc diagnosis

S5.0.0.1 Fixed thresholds (first preregistration). A threshold is the largest score at which the weighted failure rate of all calibration cases with a lower score is at most 5%. Mondrian calibration does this within the neighbour layer, the air layer and the remaining layers. Covariate-shift weights are the odds of a classifier that separates the held-out region's cases from the calibration cases on decision-time features. Local recalibration draws k independence groups from the held-out region, either alone or added to the source with up-weighting, and evaluates the remaining groups; it is repeated 30 times. The random-effects model treats the logit of each region's realised risk as normal with a region-level variance and uses an effective sample size capped at 30 cases per independence group. Table S9 gives the regional results.

S5.0.0.2 First test. The Australian roster, coordinates, network assignment and air forcing were prepared without computing any water-temperature statistic

beyond completeness counts. The models, thresholds, estimator selection, preregistration and all code used by the evaluation were frozen before it first ran. Table S10 reports the outcome of the recomputation (Text S6). Written after this evaluation, the post hoc analyses compare recoverability parameters, gap calendars, failure rates and score calibration between the source and Australia (Tables S11 and S17).

S5.0.0.3 Label-free rule. For a set of cases with predicted failure probabilities $\hat{p}_i$ sorted in increasing order, the rule admits the largest prefix whose mean of $k\hat{p}_i$ is at most 5%; tied scores are admitted together. In development, the rule was applied separately in each held-out region and population. The factor for a held-out region is the 80th percentile, over the other regions, of the ratio between realised and mean predicted risk among the cases admitted with $k = 1$, and at least one; the frozen factor uses all regions. Table S12 lists every region and the post hoc results in Australia under the corrected definitions, where the candidate models that were later frozen were applied to the opened Australian cases.

S5.0.0.4 Second test. The candidates were the 155 USGS stations that an earlier metadata intake had retained because no record of them had been cached or analysed and no known network connection linked them to analysed stations. For each, only the dates of approved daily means and their approval status were requested from the USGS daily collection, with the request fields restricted to time and approval status. Complete years (every day approved), the active span, the share of incomplete active years and their gap calendars were derived from these dates. Eligible targets have at least four complete years (71 stations); every candidate in the same eight-digit hydrologic unit as a target is a member of its network and a potential donor. Independence groups join stations that share a hydrologic unit or an intake network component. The missingness stratum is high when at least half of the active years are incomplete. NASA POWER air temperature was retrieved for the targets with the same request as for the source. The second preregistration was written and a copy was saved, and the configuration, the models, the conformal offsets, the rule and its factor, the design roster, the air forcing, the two versions of the results section and all code were frozen; only then were the approved daily values of all members requested, quality-controlled with the repository's ingest checks and arranged by network. Calendar cases of a target draw from its own in-

complete years, then from its network's, then from all new stations. Table S20 gives the results of the recomputation (Text S6). The local calibration analysis draws $m = 2, 4, 8, 16$ labelled groups, re-estimates k from them as above, applies the rule to the remaining groups, and repeats 500 times.

S6 Preregistrations, freeze and deviations

S6.0.0.1 First preregistration. It was written after the missingness audit and before any estimator comparison. It fixed the populations, independent units, primary target and tolerances, four primary endpoints with success criteria, Holm control, the decision gates and the test region. The machine-readable file was written at 21:00:09 UTC on 21 October 2026 and the document at 21:00:56; the first full case table was written at 21:04:29 and the first failure rate of the deployment population was computed at 21:05:17. Before the preregistration, a one-station case table (240 cases) was inspected once to check the code; its mean signed errors by design type were printed, but no failure rate. The document was changed once, at 21:23:45 and before the downstream experiment, by appending the clarification of P4 below. Its as-written version is archived.

S6.0.0.2 Second preregistration. It was written after the source development of the corrected definitions, the interval calibration and the label-free rule, after the Australian development check, and after the availability-only design of the second test; no water-temperature value of the new stations had been requested. A copy was saved at the moment of writing, and the pipeline was frozen seconds later (07:39:19 and 07:39:22 UTC on 22 October 2026).

S6.0.0.3 Deviations.

1. Before the downstream experiment ran, the coverage criterion of P4 was clarified to refer to the missing-data component of the multiple-imputation interval around the complete-record slope.
2. P3 requires success in both populations, the stricter reading of "mech + deploy".

3. The recovery law gained the window-scale version (v2) and the replay (v3) during development; v1 is reported. All three were fixed before the first freeze; v2p was added in the revision.
4. A first run of the downstream experiment used large-sample Rubin degrees of freedom; the reported results use Barnard–Rubin degrees of freedom.
5. The label-free threshold was formulated after the first test; it was developed on the source and evaluated in the second test under the second preregistration. Australian results under this rule are post hoc.
6. Return levels, a cause-specific missingness analysis and a natural-gap comparison with independent reference sensors were not possible with the available data.
7. The first frozen pipeline used calendar years and a May–September warm season, which is not adapted to the Southern Hemisphere. In the revision, all analyses use hot years and station seasons; the verdicts of P1–P4 are those of the registered analysis, and the corrected values are reported beside them.
8. After the first preregistration and before the first freeze, the case builder gained outputs for the ablations (unshrunk air regression, the earlier peak-window summaries, training-year replay errors) and a mode for real incomplete years. All previously existing columns were verified to be identical after the rebuild. The "case table at the time of preregistration" in the first version of this deviation is the first full table, completed five minutes after the preregistration.
9. Gate C anticipated a short note if P2 failed. P2 failed because D2* without screening already met the 5% risk under real calendars, which makes the gain at 5% uninformative rather than negative; the main claims rest on P1, P4 and the screening of long gaps, so the study is reported as a full article.
10. The interval calibration by difficulty group was added after P1 had been evaluated; P1 is judged on the registered uncalibrated intervals.
11. The peak-season rules are anchored to each station's climatological peak in the revision; the rules as written are reported in Table 2 of the main text.
12. The low-flow demonstration and the decomposition of the downstream gain were added in the revision and are exploratory.

13. After both tests had been evaluated, the generation of random numbers was changed so that every seed is a plain integer (a fixed number, or one formed from the position of a station or region in a sorted list, the year and the step of the design). This redrew every mask, calendar draw, shift, simulation and resample. All results were recomputed with otherwise unchanged definitions, models, rules and code; the machine-readable preregistration files were regenerated by the same code, with a new creation time. The original evaluations, which can no longer be reproduced exactly, gave: P1 a reduction of 73% with 85.1% coverage (met); P2 a gain of 0.0 points (not met); P3 86% of regions under long gaps (median 5.7%), 93% under real calendars, and 12.6% and 17.5% realised risk in Australia (not met); P4 a reduction of 76% with 95.9% coverage (met); N1 4.4% (upper bound 6.2%) under real calendars and 5.1% (6.3%) under long gaps with $k = 1.27$ (met); N2 87.9% coverage (met); N3 a reduction of 88% (met); N4 gains of 44 and 2 points. In the recomputation P1 is not met, because its coverage of 84.8% fell just below the registered floor of 85%, and P3 is not met with 79% of regions under long gaps (median 5.6%) and Australian risks of 13.1% and 17.6%; P4 is met with a reduction of 65% (67% under the corrected definitions); N1–N3 are met (4.5% and 5.6% realised risk with $k = 1.27$, 87.4% coverage, a bias reduction of 86%), and local re-estimation of k needed eight labelled groups under long gaps instead of two. The main text and the tables report the recomputed values. Because the change was made after both outcomes were known, neither test is reported as blind.

S7 Downstream decomposition and the 7-day minimum flow

S7.0.0.1 Decomposition. For each station and pipeline, the trend RMSE is computed over the 200 degradations. The share of the A1-to-C reduction obtained by keeping years is $(\mathrm{RMSE}_{A1} - \mathrm{RMSE}_{B})/(\mathrm{RMSE}_{A1} - \mathrm{RMSE}_{C})$ over all stations, with an interval from resampling independence groups. Station ratios are undefined where the denominator pipeline is exact; at 10 stations no transplanted calendar reached the peak window and the plug-in was exact. Detection of warming uses the t interval of each pipeline (Barnard–Rubin degrees of freedom for C). The Baden-Württemberg trends use the 19 stations with at least five complete years and at

least 15 years in total, and the same 20 multiple imputations per incomplete year as in the main analysis; the anomaly series are the year effects of a least-squares model with station and year effects, centred on zero (Table S21).

S7.0.0.2 Low flow. Discharge values with an approval code (including approved estimates) are valid; streams with any zero flow are excluded because the log error of 7Q is undefined when 7Q is zero. A climatic year runs from 1 April to 31 March and is complete when every day is valid. The climatology of log flow is a 31-day circular moving average of the training years' day-of-year means, and the low-flow window is the 30 days centred on the median position of the training years' 7Q. Controlled gaps of 30, 60 and 90 days have no, partial or full overlap with this window (one per cell); real calendars come from the 155 interior incomplete climatic years with at least 30 valid days, moved by up to 15 days, six per complete year. Neighbours lie within 200 km, have an anomaly correlation of at least 0.6 over at least three years of common training days and are complete in the target year; at most three are used in a ridge regression with unit penalty. The target-aware estimator predicts seven quantiles of the log error with the same LightGBM settings as P5a, from the missing fraction, the number and longest length of gaps, the overlap with the low-flow window, gaps at the year boundary, the layer, the number of neighbours and the regression R^2, the plug-in 7Q relative to the training mean, the recorded-day 7Q relative to the plug-in, the best neighbour's 7Q anomaly in the target year, the training standard deviation of log 7Q, the distance of the gap centre from the low-flow window and the visible anomaly next to the gaps. The trend experiment degrades each station with at least 20 complete climatic years 200 times at random, with the station's real share of incomplete years (between 5% and 60%), and estimates the OLS trend of log 7Q per decade. Table S22 gives the results.

Table S1. E1 missingness models. Station-year models use station fixed effects (conditional logit) or station and region-year random intercepts; onset models are time-stratified case-crossover designs (strata: station × year × month). Duration is a ratio of gap lengths. Intervals resample independence groups (model-based for the mixed model and duration).

Analysis	Covariate	n	Odds ratio [95%]
station year conditional logit	z air	8,848	1.06 [1.00, 1.13]
station year conditional logit	trend	8,848	0.68 [0.59, 0.81]
station year conditional logit below90	z air	6,275	1.03 [0.95, 1.11]
station year conditional logit below90	trend	6,275	0.62 [0.50, 0.76]
station year conditional logit julaug	z air	8,148	1.06 [0.99, 1.13]
station year conditional logit julaug	trend	8,148	0.68 [0.59, 0.80]
station year conditional logit usgs flow	z air	4,632	1.02 [0.94, 1.12]
station year conditional logit usgs flow	z flow	4,632	0.94 [0.87, 1.01]
station year conditional logit usgs flow	trend	4,632	0.66 [0.54, 0.81]
station year mixed station region year	z air	10,580	1.13 [1.07, 1.18]
station year mixed station region year	trend	10,580	0.59 [0.57, 0.61]
onset case crossover air	z air week	105,238	1.07 [1.01, 1.12]
onset case crossover air water	z air week	105,238	1.07 [1.02, 1.11]
onset case crossover air water	z water lag	105,238	1.01 [0.96, 1.07]
onset case crossover usgs flow	z air week	61,052	1.06 [0.99, 1.12]
onset case crossover usgs flow	low flow	61,052	1.17 [0.93, 1.55]
gap log duration station fe	z air week	5,392	0.98 [0.94, 1.01]

Table S2. Target tolerances: the share of the median within-station interannual standard deviation that 0.5 °C represents for H30, with floors of 0.2 °C and one day, recomputed for the degree-days of each station's four climatologically hottest months.

Target	Median interannual SD	Tolerance
Annual mean	0.53	0.26
C30	0.80	0.40
Days >20 °C	10.95	5.45
Degree-days, 4 hottest months	94.36	46.97
H30	1.00	0.50
H7	1.07	0.53

References

Romano, Y., Patterson, E., & Candès, E. J. (2019). Conformalized quantile regression. In *Advances in neural information processing systems* (Vol. 32).

Table S3. E3 recovery law, case level. v1: daily-scale shrinkage; v2: window-scale shrinkage (main parametric law); v2p: residual moments inside the training peak windows; v3: replay of the same gap on training years (empirical comparator). No version is fitted to reconstruction errors.

Population	Law	ECE	Brier skill	AUC	Predicted failure	Observed failure	Predicted bias	Observed bias
deploy	v1	0.002	0.426	0.978	0.032	0.033	−0.038	−0.026
deploy	v2	0.002	0.428	0.978	0.032	0.033	−0.036	−0.026
deploy	v2p	0.006	0.418	0.979	0.037	0.033	−0.051	−0.026
deploy	v3	0.009	0.412	0.978	0.037	0.032	−0.027	−0.026
mech	v1	0.023	0.266	0.866	0.148	0.170	−0.147	−0.064
mech	v2	0.027	0.266	0.867	0.143	0.170	−0.133	−0.064
mech	v2p	0.027	0.274	0.869	0.165	0.170	−0.187	−0.064
mech	v3	0.039	0.256	0.863	0.182	0.171	−0.063	−0.068

Table S4. E3 cells of the controlled population (gap length × overlap with the climatological peak × information layer).

Days	Overlap	Layer	n	Mean error	v2 bias	v3 bias	Failure	v2 P(fail)	v2p P(fail)	v3 P(fail)
30	full	air ridge	1,964	−0.290	−0.331	−0.293	0.401	0.341	0.428	0.432
30	full	donor ridge	10,422	−0.002	−0.134	−0.020	0.185	0.152	0.175	0.210
30	none	air ridge	1,742	−0.002	−0.014	−0.003	0.013	0.009	0.011	0.010
30	none	donor ridge	10,670	0.001	−0.009	0.002	0.006	0.004	0.005	0.006
30	partial	air ridge	1,909	−0.185	−0.259	−0.153	0.260	0.237	0.307	0.295
30	partial	donor ridge	10,495	−0.006	−0.119	−0.004	0.132	0.104	0.121	0.143
60	full	air ridge	3,119	−0.440	−0.386	−0.417	0.519	0.459	0.528	0.537
60	full	donor ridge	9,319	−0.055	−0.167	−0.056	0.281	0.229	0.254	0.302
60	full	smooth climatology	114	−0.435	−0.766	−0.399	0.658	0.672	0.717	0.674
60	none	air ridge	2,679	−0.011	−0.025	−0.002	0.018	0.015	0.018	0.016
60	none	donor ridge	9,760	0.001	−0.012	0.001	0.008	0.006	0.007	0.007
60	none	smooth climatology	117	−0.016	−0.032	−0.014	0.009	0.019	0.022	0.012
60	partial	air ridge	2,979	−0.208	−0.286	−0.180	0.286	0.261	0.328	0.313
60	partial	donor ridge	9,451	−0.011	−0.137	−0.011	0.153	0.124	0.143	0.167
60	partial	smooth climatology	118	−0.163	−0.538	−0.207	0.364	0.403	0.476	0.460
90	full	air ridge	3,869	−0.500	−0.402	−0.470	0.552	0.499	0.565	0.570
90	full	donor ridge	8,596	−0.073	−0.176	−0.074	0.312	0.256	0.282	0.335
90	full	smooth climatology	118	−0.534	−0.850	−0.464	0.695	0.708	0.742	0.700
90	none	air ridge	3,523	−0.014	−0.032	−0.008	0.023	0.020	0.023	0.020
90	none	donor ridge	8,937	0.001	−0.017	0.002	0.011	0.008	0.009	0.011
90	none	smooth climatology	122	−0.027	−0.031	−0.015	0.025	0.019	0.021	0.011
90	partial	air ridge	3,746	−0.204	−0.288	−0.179	0.294	0.260	0.326	0.307
90	partial	donor ridge	8,703	−0.011	−0.144	−0.015	0.161	0.128	0.148	0.171
90	partial	smooth climatology	121	−0.203	−0.499	−0.185	0.421	0.372	0.453	0.428

Table S5. E4 H30 estimators, leave-one-region-out, before interval calibration. Strata use the true peak window (evaluation only); warm anomaly: true H30 at least 1 °C above the training mean. Calibrated coverage by difficulty group is in Table S19.

Population	Stratum	Estimator	n	Bias	MAE	Failure (%)	CRPS	Cover90 (%)
mech	all	P0 plug-in	112,593	−0.064	0.260	17.0	–	–
mech	all	Recorded days only	112,593	−0.781	0.802	38.7	–	–
mech	all	Air2stream-3	110,609	−0.090	0.325	23.4	–	–
mech	all	P2 functional debiasing	112,593	0.004	0.209	12.7	–	–
mech	all	P3 air-state simulation	112,593	−0.011	0.223	14.3	0.134	88.3
mech	all	P3 raw residual blocks	112,593	−0.024	0.254	16.7	0.162	81.1
mech	all	P5a distribution regression	112,593	−0.019	0.203	12.6	0.122	90.0
mech	all	P5b = D2*	112,593	−0.016	0.197	12.0	0.118	90.2
mech	peak half hidden	P0 plug-in	49,935	−0.174	0.493	33.9	–	–
mech	peak half hidden	Recorded days only	49,935	−1.712	1.714	82.1	–	–
mech	peak half hidden	Air2stream-3	49,072	−0.230	0.626	47.0	–	–
mech	peak half hidden	P2 functional debiasing	49,935	−0.059	0.375	24.9	–	–
mech	peak half hidden	P3 air-state simulation	49,935	−0.060	0.417	28.2	0.249	81.2
mech	peak half hidden	P3 raw residual blocks	49,935	−0.096	0.473	32.9	0.301	68.4
mech	peak half hidden	P5a distribution regression	49,935	−0.089	0.380	25.4	0.227	83.3
mech	peak half hidden	P5b = D2*	49,935	−0.080	0.370	24.2	0.220	84.0
mech	air layer half hidden	P0 plug-in	12,325	−0.473	0.732	52.4	–	–
mech	air layer half hidden	Recorded days only	12,325	−1.861	1.863	84.6	–	–
mech	air layer half hidden	Air2stream-3	12,325	−0.250	0.634	46.9	–	–
mech	air layer half hidden	P2 functional debiasing	12,325	−0.096	0.460	33.1	–	–
mech	air layer half hidden	P3 air-state simulation	12,325	−0.171	0.510	37.0	0.304	82.8
mech	air layer half hidden	P3 raw residual blocks	12,325	−0.329	0.657	48.8	0.407	70.1
mech	air layer half hidden	P5a distribution regression	12,325	−0.144	0.473	34.2	0.281	84.3
mech	air layer half hidden	P5b = D2*	12,325	−0.137	0.458	32.3	0.272	84.8
mech	donor layer half hidden	P0 plug-in	37,271	−0.072	0.410	27.5	–	–
mech	donor layer half hidden	Recorded days only	37,271	−1.654	1.657	81.3	–	–
mech	donor layer half hidden	Air2stream-3	36,747	−0.223	0.623	47.1	–	–
mech	donor layer half hidden	P2 functional debiasing	37,271	−0.046	0.343	21.8	–	–
mech	donor layer half hidden	P3 air-state simulation	37,271	−0.022	0.382	25.0	0.228	80.7
mech	donor layer half hidden	P3 raw residual blocks	37,271	−0.017	0.408	27.3	0.264	67.8
mech	donor layer half hidden	P5a distribution regression	37,271	−0.069	0.345	22.2	0.206	83.0
mech	donor layer half hidden	P5b = D2*	37,271	−0.060	0.337	21.2	0.200	83.8
mech	warm anomaly half hidden	P0 plug-in	9,995	−0.623	0.727	50.1	–	–
mech	warm anomaly half hidden	Recorded days only	9,995	−2.262	2.265	90.9	–	–
mech	warm anomaly half hidden	Air2stream-3	9,833	−0.717	0.893	64.8	–	–
mech	warm anomaly half hidden	P2 functional debiasing	9,995	−0.387	0.513	36.7	–	–
mech	warm anomaly half hidden	P3 air-state simulation	9,995	−0.472	0.603	43.8	0.365	67.3
mech	warm anomaly half hidden	P3 raw residual blocks	9,995	−0.566	0.681	48.8	0.463	51.1
mech	warm anomaly half hidden	P5a distribution regression	9,995	−0.424	0.538	38.5	0.328	73.1
mech	warm anomaly half hidden	P5b = D2*	9,995	−0.402	0.524	37.4	0.318	73.4
deploy	all	P0 plug-in	31,178	−0.026	0.060	3.3	–	–
deploy	all	Recorded days only	31,114	−0.313	0.330	7.0	–	–
deploy	all	Air2stream-3	26,176	−0.013	0.041	2.2	–	–
deploy	all	P2 functional debiasing	31,178	0.000	0.051	2.7	–	–
deploy	all	P3 air-state simulation	31,178	−0.007	0.050	2.8	0.029	94.9
deploy	all	P3 raw residual blocks	31,178	−0.013	0.057	3.0	0.035	93.7
deploy	all	P5a distribution regression	31,178	−0.002	0.045	2.5	0.027	97.2
deploy	all	P5b = D2*	31,178	−0.003	0.044	2.4	0.026	97.3
deploy	peak half hidden	P0 plug-in	2,592	−0.314	0.627	42.4	–	–
deploy	peak half hidden	Recorded days only	2,538	−4.546	4.547	89.2	–	–
deploy	peak half hidden	Air2stream-3	1,509	−0.217	0.612	46.5	–	–
deploy	peak half hidden	P2 functional debiasing	2,592	−0.040	0.483	34.5	–	–
deploy	peak half hidden	P3 air-state simulation	2,592	−0.080	0.496	37.5	0.292	81.7
deploy	peak half hidden	P3 raw residual blocks	2,592	−0.150	0.576	39.3	0.364	68.3
deploy	peak half hidden	P5a distribution regression	2,592	−0.059	0.482	34.4	0.285	85.0
deploy	peak half hidden	P5b = D2*	2,592	−0.064	0.470	33.5	0.274	85.5
deploy	air layer half hidden	P0 plug-in	828	−0.489	0.778	53.3	–	–
deploy	air layer half hidden	Recorded days only	812	−5.375	5.376	92.5	–	–
deploy	air layer half hidden	Air2stream-3	424	−0.177	0.646	49.5	–	–
deploy	air layer half hidden	P2 functional debiasing	828	−0.051	0.533	39.0	–	–
deploy	air layer half hidden	P3 air-state simulation	828	−0.122	0.544	42.9	0.320	82.2
deploy	air layer half hidden	P3 raw residual blocks	828	−0.260	0.686	47.2	0.430	69.0
deploy	air layer half hidden	P5a distribution regression	828	−0.079	0.541	39.5	0.319	86.4
deploy	air layer half hidden	P5b = D2*	828	−0.091	0.522	38.7	0.304	86.0
deploy	donor layer half hidden	P0 plug-in	1,712	−0.113	0.446	28.7	–	–
deploy	donor layer half hidden	Recorded days only	1,674	−3.597	3.597	85.2	–	–
deploy	donor layer half hidden	Air2stream-3	1,083	−0.254	0.582	43.7	–	–
deploy	donor layer half hidden	P2 functional debiasing	1,712	−0.034	0.414	27.9	–	–
deploy	donor layer half hidden	P3 air-state simulation	1,712	−0.035	0.426	29.9	0.252	80.9
deploy	donor layer half hidden	P3 raw residual blocks	1,712	−0.033	0.439	29.2	0.283	67.3
deploy	donor layer half hidden	P5a distribution regression	1,712	−0.038	0.404	27.3	0.241	83.9
deploy	donor layer half hidden	P5b = D2*	1,712	−0.038	0.399	26.4	0.234	84.9
deploy	warm anomaly half hidden	P0 plug-in	530	−1.039	1.062	63.8	–	–
deploy	warm anomaly half hidden	Recorded days only	519	−5.040	5.040	95.4	–	–
deploy	warm anomaly half hidden	Air2stream-3	317	−0.767	0.895	61.1	–	–
deploy	warm anomaly half hidden	P2 functional debiasing	530	−0.672	0.727	54.7	–	–
deploy	warm anomaly half hidden	P3 air-state simulation	530	−0.773	0.813	58.8	0.488	62.6
deploy	warm anomaly half hidden	P3 raw residual blocks	530	−0.911	0.941	60.5	0.648	44.6
deploy	warm anomaly half hidden	P5a distribution regression	530	−0.709	0.754	54.2	0.457	68.5
deploy	warm anomaly half hidden	P5b = D2*	530	−0.698	0.751	53.5	0.446	68.4

Table S6. E4 estimators for the other targets (all cases), with the target tolerances.

Population	Target	Estimator	Bias	MAE	Failure (%)
mech	H7	P0 plug-in	−0.100	0.284	18.0
mech	H7	Recorded days only	−0.647	0.651	31.7
mech	H7	P2 functional debiasing	0.013	0.247	14.7
mech	H7	P3 air-state simulation	0.001	0.252	16.6
mech	H7	P5b = D2*	−0.033	0.228	14.4
mech	Annual mean	P0 plug-in	−0.003	0.076	4.8
mech	Annual mean	Recorded days only	−0.836	1.134	90.3
mech	Annual mean	P2 functional debiasing	0.001	0.062	2.5
mech	Annual mean	P3 air-state simulation	−0.000	0.067	3.5
mech	Annual mean	P5b = D2*	0.001	0.058	2.2
mech	C30	P0 plug-in	−0.006	0.022	1.7
mech	C30	Recorded days only	0.019	0.102	6.4
mech	C30	P2 functional debiasing	0.000	0.017	1.2
mech	C30	P3 air-state simulation	−0.004	0.019	1.5
mech	C30	P5b = D2*	0.002	0.015	1.1
mech	Degree-days, 4 hottest months	P0 plug-in	−1.826	20.548	12.4
mech	Degree-days, 4 hottest months	Recorded days only	−42.168	70.208	50.0
mech	Degree-days, 4 hottest months	P2 functional debiasing	0.198	16.659	8.2
mech	Degree-days, 4 hottest months	P3 air-state simulation	−0.040	17.950	9.9
mech	Degree-days, 4 hottest months	P5b = D2*	0.233	15.700	7.6
mech	Days $>20\,^{\circ}$C	P0 plug-in	−0.182	1.992	11.4
mech	Days $>20\,^{\circ}$C	Recorded days only	−4.237	6.753	34.7
mech	Days $>20\,^{\circ}$C	P2 functional debiasing	−0.027	1.851	10.0
mech	Days $>20\,^{\circ}$C	P3 air-state simulation	0.048	1.751	9.9
mech	Days $>20\,^{\circ}$C	P5b = D2*	−0.009	1.635	9.2
deploy	H7	P0 plug-in	−0.034	0.065	3.8
deploy	H7	Recorded days only	−0.271	0.275	6.1
deploy	H7	P2 functional debiasing	0.004	0.061	3.1
deploy	H7	P3 air-state simulation	−0.004	0.054	3.4
deploy	H7	P5b = D2*	−0.007	0.049	2.9
deploy	Annual mean	P0 plug-in	0.001	0.036	2.7
deploy	Annual mean	Recorded days only	0.105	0.507	30.3
deploy	Annual mean	P2 functional debiasing	0.001	0.032	2.0
deploy	Annual mean	P3 air-state simulation	0.001	0.032	2.2
deploy	Annual mean	P5b = D2*	0.000	0.029	1.9
deploy	C30	P0 plug-in	−0.017	0.080	5.9
deploy	C30	Recorded days only	0.293	0.386	14.3
deploy	C30	P2 functional debiasing	0.001	0.064	5.0
deploy	C30	P3 air-state simulation	−0.011	0.065	4.9
deploy	C30	P5b = D2*	0.008	0.057	4.4
deploy	Degree-days, 4 hottest months	P0 plug-in	−0.398	5.701	3.0
deploy	Degree-days, 4 hottest months	Recorded days only	−5.500	14.437	6.7
deploy	Degree-days, 4 hottest months	P2 functional debiasing	0.110	5.217	2.4
deploy	Degree-days, 4 hottest months	P3 air-state simulation	0.147	5.109	2.4
deploy	Degree-days, 4 hottest months	P5b = D2*	0.312	4.500	2.2
deploy	Days $>20\,^{\circ}$C	P0 plug-in	−0.042	0.646	3.2
deploy	Days $>20\,^{\circ}$C	Recorded days only	−1.206	2.000	7.9
deploy	Days $>20\,^{\circ}$C	P2 functional debiasing	0.044	0.669	2.8
deploy	Days $>20\,^{\circ}$C	P3 air-state simulation	0.063	0.568	2.7
deploy	Days $>20\,^{\circ}$C	P5b = D2*	0.045	0.529	2.4

Table S7. E5 scores: retention at equal realised risk, AURC and AUGRC for each delivered value (D2* with calibrated intervals).

Population	Value	Score	2.5%	5%	10%	AURC	AUGRC
deploy	P0	Annual missing fraction (threshold varied)	98.2	100.0	100.0	0.0032	0.0028
deploy	P0	Warm-season missing fraction (threshold varied)	98.4	100.0	100.0	0.0017	0.0016
deploy	P0	Observed lower bound (NRFA type)	98.0	100.0	100.0	0.0071	0.0048
deploy	P0	Daily reconstruction disagreement	0.2	100.0	100.0	0.0353	0.0169
deploy	P0	H30 reconstruction disagreement	98.4	100.0	100.0	0.0024	0.0020
deploy	P0	Recovery law, parametric (v2)	98.9	100.0	100.0	0.0014	0.0012
deploy	P0	Recovery law, training-year replay (v3)	95.7	96.7	96.7	0.0013	0.0011
deploy	P0	Target-aware: D2* predictive distribution	99.0	100.0	100.0	0.0025	0.0017
deploy	P0	Direct failure probability (plug-in value)	99.0	100.0	100.0	0.0009	0.0009
deploy	P0	Direct failure probability (D2* value)	98.9	100.0	100.0	0.0010	0.0010
deploy	P0	Oracle error order (bound)	99.2	100.0	100.0	0.0005	0.0005
deploy	Pobs	Annual missing fraction (threshold varied)	87.7	97.8	100.0	0.0082	0.0072
deploy	Pobs	Warm-season missing fraction (threshold varied)	94.3	97.8	100.0	0.0042	0.0040
deploy	Pobs	Observed lower bound (NRFA type)	88.8	97.4	100.0	0.0124	0.0091
deploy	Pobs	Daily reconstruction disagreement	0.2	0.3	100.0	0.0725	0.0362
deploy	Pobs	H30 reconstruction disagreement	94.0	97.7	100.0	0.0046	0.0041
deploy	Pobs	Recovery law, parametric (v2)	93.9	97.7	100.0	0.0047	0.0043
deploy	Pobs	Recovery law, training-year replay (v3)	91.1	94.7	96.7	0.0052	0.0040
deploy	Pobs	Target-aware: D2* predictive distribution	94.8	97.7	100.0	0.0145	0.0082
deploy	Pobs	Direct failure probability (plug-in value)	94.6	97.6	100.0	0.0036	0.0034
deploy	Pobs	Direct failure probability (D2* value)	94.8	97.7	100.0	0.0034	0.0033
deploy	Pobs	Oracle error order (bound)	95.3	97.9	100.0	0.0025	0.0025
deploy	D2*	Annual missing fraction (threshold varied)	100.0	100.0	100.0	0.0019	0.0017
deploy	D2*	Warm-season missing fraction (threshold varied)	100.0	100.0	100.0	0.0010	0.0010
deploy	D2*	Observed lower bound (NRFA type)	100.0	100.0	100.0	0.0043	0.0031
deploy	D2*	Daily reconstruction disagreement	100.0	100.0	100.0	0.0271	0.0128
deploy	D2*	H30 reconstruction disagreement	100.0	100.0	100.0	0.0018	0.0015
deploy	D2*	Recovery law, parametric (v2)	100.0	100.0	100.0	0.0008	0.0008
deploy	D2*	Recovery law, training-year replay (v3)	96.7	96.7	96.7	0.0008	0.0007
deploy	D2*	Target-aware: D2* predictive distribution	100.0	100.0	100.0	0.0012	0.0009
deploy	D2*	Direct failure probability (plug-in value)	100.0	100.0	100.0	0.0007	0.0007
deploy	D2*	Direct failure probability (D2* value)	100.0	100.0	100.0	0.0007	0.0006
deploy	D2*	Oracle error order (bound)	100.0	100.0	100.0	0.0003	0.0003
mech	P0	Annual missing fraction (threshold varied)	0.0	0.0	0.0	0.1413	0.0760
mech	P0	Warm-season missing fraction (threshold varied)	22.8	22.8	57.3	0.0743	0.0543
mech	P0	Observed lower bound (NRFA type)	0.1	20.3	63.3	0.0889	0.0555
mech	P0	Daily reconstruction disagreement	0.3	1.3	8.9	0.1333	0.0725
mech	P0	H30 reconstruction disagreement	47.6	57.7	75.9	0.0525	0.0407
mech	P0	Recovery law, parametric (v2)	53.8	66.9	84.4	0.0413	0.0332
mech	P0	Recovery law, training-year replay (v3)	50.0	63.1	81.9	0.0417	0.0319
mech	P0	Target-aware: D2* predictive distribution	64.6	74.2	87.1	0.0389	0.0293
mech	P0	Direct failure probability (plug-in value)	67.0	75.9	88.3	0.0301	0.0258
mech	P0	Direct failure probability (D2* value)	65.1	73.9	86.5	0.0326	0.0279
mech	P0	Oracle error order (bound)	85.2	87.4	92.3	0.0153	0.0144
mech	Pobs	Annual missing fraction (threshold varied)	0.0	0.0	0.0	0.3431	0.1788
mech	Pobs	Warm-season missing fraction (threshold varied)	22.8	22.8	22.8	0.1719	0.1254
mech	Pobs	Observed lower bound (NRFA type)	0.1	9.9	28.8	0.1753	0.1165
mech	Pobs	Daily reconstruction disagreement	0.2	0.2	0.4	0.3270	0.1751
mech	Pobs	H30 reconstruction disagreement	44.1	49.2	56.8	0.1185	0.0941
mech	Pobs	Recovery law, parametric (v2)	28.1	30.1	36.0	0.1662	0.1199
mech	Pobs	Recovery law, training-year replay (v3)	29.0	32.0	38.4	0.1618	0.1117
mech	Pobs	Target-aware: D2* predictive distribution	0.0	0.0	0.0	0.1850	0.1119
mech	Pobs	Direct failure probability (plug-in value)	35.4	40.6	51.4	0.1292	0.1004
mech	Pobs	Direct failure probability (D2* value)	36.3	40.8	52.3	0.1256	0.0979
mech	Pobs	Oracle error order (bound)	62.9	64.6	68.1	0.0869	0.0748
mech	D2*	Annual missing fraction (threshold varied)	0.0	0.0	41.9	0.0909	0.0509
mech	D2*	Warm-season missing fraction (threshold varied)	22.8	56.3	88.1	0.0482	0.0358
mech	D2*	Observed lower bound (NRFA type)	11.6	47.6	88.9	0.0582	0.0368
mech	D2*	Daily reconstruction disagreement	0.8	4.5	78.0	0.0857	0.0480
mech	D2*	H30 reconstruction disagreement	53.1	68.6	93.4	0.0350	0.0269
mech	D2*	Recovery law, parametric (v2)	58.8	74.6	94.8	0.0294	0.0236
mech	D2*	Recovery law, training-year replay (v3)	55.3	71.0	91.5	0.0301	0.0231
mech	D2*	Target-aware: D2* predictive distribution	69.9	80.9	95.9	0.0264	0.0199
mech	D2*	Direct failure probability (plug-in value)	69.2	79.8	95.0	0.0230	0.0196
mech	D2*	Direct failure probability (D2* value)	70.0	80.9	95.9	0.0219	0.0188
mech	D2*	Oracle error order (bound)	90.2	92.6	97.7	0.0076	0.0072

Table S8. E6 agreement of usability between targets for the plug-in value.

Population	Target A	Target B	κ	Spearman	A no, B yes (%)	A yes, B no (%)
deploy	H30	H7	0.68	0.66	0.8	1.3
deploy	H30	Annual mean	0.37	0.33	2.1	1.5
deploy	H30	C30	0.18	0.14	2.3	4.9
deploy	H30	Degree-days, 4 hottest months	0.62	0.56	1.3	1.0
deploy	H30	Days $>20\,^{\circ}$C	0.36	0.34	2.0	1.9
deploy	H7	Annual mean	0.31	0.34	2.7	1.6
deploy	H7	C30	0.18	0.16	2.8	4.8
deploy	H7	Degree-days, 4 hottest months	0.51	0.48	2.0	1.2
deploy	H7	Days $>20\,^{\circ}$C	0.35	0.34	2.5	1.9
deploy	Annual mean	C30	0.27	0.34	1.4	4.6
deploy	Annual mean	Degree-days, 4 hottest months	0.52	0.47	1.2	1.5
deploy	Annual mean	Days $>20\,^{\circ}$C	0.30	0.34	1.8	2.2
deploy	C30	Degree-days, 4 hottest months	0.17	0.04	4.9	2.1
deploy	C30	Days $>20\,^{\circ}$C	0.16	0.11	5.0	2.3
deploy	Degree-days, 4 hottest months	Days $>20\,^{\circ}$C	0.40	0.45	1.7	1.9
mech	H30	H7	0.64	0.79	4.7	5.7
mech	H30	Annual mean	0.21	0.35	14.1	1.9
mech	H30	C30	−0.03	−0.28	17.0	1.7
mech	H30	Degree-days, 4 hottest months	0.51	0.68	8.4	3.9
mech	H30	Days $>20\,^{\circ}$C	0.26	0.30	11.8	6.2
mech	H7	Annual mean	0.16	0.25	15.4	2.2
mech	H7	C30	−0.03	−0.24	18.0	1.7
mech	H7	Degree-days, 4 hottest months	0.38	0.54	10.8	5.2
mech	H7	Days $>20\,^{\circ}$C	0.22	0.26	13.1	6.5
mech	Annual mean	C30	0.02	−0.01	4.6	1.6
mech	Annual mean	Degree-days, 4 hottest months	0.41	0.65	0.9	8.5
mech	Annual mean	Days $>20\,^{\circ}$C	0.19	0.23	2.8	9.4
mech	C30	Degree-days, 4 hottest months	−0.03	−0.37	1.7	12.4
mech	C30	Days $>20\,^{\circ}$C	−0.03	−0.18	1.7	11.4
mech	Degree-days, 4 hottest months	Days $>20\,^{\circ}$C	0.33	0.39	7.5	6.5

Table S9. E7 (first preregistration, P3) leave-one-region-out realised risk (%) of the fixed thresholds at nominal 5%: one threshold, one per information layer, and layer thresholds reweighted to the held-out region.

Population	Region	Groups	Marginal	Layer	Weighted
deploy	arso	12	3.1	3.0	3.0
deploy	chmi	17	7.1	7.1	7.1
deploy	gkd bayern	9	1.1	0.9	0.9
deploy	lubw	7	1.5	0.9	0.9
deploy	nve hydapi	17	7.3	5.7	5.3
deploy	usgs huc02	11	3.1	1.6	1.6
deploy	usgs huc03	15	1.6	1.2	1.1
deploy	usgs huc04	9	0.6	0.4	0.4
deploy	usgs huc05	11	1.9	1.0	1.0
deploy	usgs huc07	5	3.0	3.0	3.0
deploy	usgs huc10	7	2.9	2.1	2.0
deploy	usgs huc17	18	2.1	1.6	1.4
deploy	usgs huc18	5	2.1	1.6	1.6
deploy	usgs huc19	8	2.1	1.5	1.5
mech	arso	12	5.4	5.6	5.5
mech	chmi	17	7.0	7.7	5.7
mech	gkd bayern	9	4.1	4.0	3.8
mech	lubw	7	3.6	2.9	2.9
mech	nve hydapi	17	6.3	6.7	5.9
mech	usgs huc02	11	6.2	6.4	5.9
mech	usgs huc03	15	4.6	4.3	3.8
mech	usgs huc04	9	3.3	3.1	3.1
mech	usgs huc05	11	5.7	5.6	5.1
mech	usgs huc07	5	7.8	8.2	8.2
mech	usgs huc10	7	5.3	5.1	5.2
mech	usgs huc17	18	4.0	3.9	3.1
mech	usgs huc18	5	6.8	6.4	6.4
mech	usgs huc19	8	7.2	7.2	7.2

Table S10. E7 first test (Australia): frozen thresholds, recomputed after the change of random seeds (Text S6). Intervals resample the 55 hydrological networks.

Population	Score	Threshold	Cases	Networks	Used (%)	Risk (%)	Low	High	Unscreened (%)
deploy	R7a target aware	marginal	8150	55	100.0	14.7	12.5	16.5	14.7
deploy	R7a target aware	mondrian layer	8150	55	97.4	13.1	11.0	14.7	14.7
deploy	R7b direct D2	marginal	8150	55	100.0	14.7	12.5	16.5	14.7
deploy	R7b direct D2	mondrian layer	8150	55	97.3	13.3	11.3	14.9	14.7
deploy	R6 law	marginal	8150	55	100.0	17.1	14.9	19.1	17.1
deploy	R6 law	mondrian layer	8150	55	94.4	14.6	12.3	16.7	17.1
deploy	R7b direct P0	marginal	8150	55	100.0	17.1	14.9	19.1	17.1
deploy	R7b direct P0	mondrian layer	8150	55	95.3	14.3	12.1	16.4	17.1
mech	R7a target aware	marginal	22578	55	99.9	18.7	16.2	20.6	18.8
mech	R7a target aware	mondrian layer	22578	55	97.7	17.6	14.9	19.5	18.8
mech	R7b direct D2	marginal	22578	55	100.0	18.8	16.2	20.7	18.8
mech	R7b direct D2	mondrian layer	22578	55	98.3	17.9	15.4	19.7	18.8
mech	R6 law	marginal	22578	55	100.0	22.9	19.8	25.7	22.9
mech	R6 law	mondrian layer	22578	55	94.5	20.9	17.4	24.1	22.9
mech	R7b direct P0	marginal	22578	55	100.0	22.9	19.8	25.7	22.9
mech	R7b direct P0	mondrian layer	22578	55	93.1	19.8	15.9	23.1	22.9

Table S11. Post hoc contrast between the source and Australia, written after the first evaluation in Australia (calendar years, May–September seasons).

Domain	Family	Air (%)	σ_a	σ_r	R^2	Missing (%)	Peak ≥50% (%)	Fail P0	Fail D2*	Cover90
source	calendar	14.0	1.77	0.76	0.78	2.7	8.4	3.7	2.6	98.2
source	crossed	22.7	1.77	0.83	0.74	16.4	50.8	17.0	10.0	94.7
australia	calendar	55.9	1.39	0.93	0.52	14.2	27.6	17.1	14.7	92.4
australia	crossed	54.2	1.39	0.91	0.52	16.4	44.6	22.9	18.8	88.8

Table S12. Label-free expected-risk rule (M3), developed leave-one-region-out: admit cases by increasing score while k times the mean predicted failure probability among admitted cases stays at or below 5%. The factor k for each region is the 80th percentile of the realised-to-predicted ratio in the other regions. Risks refer to the D2* value. Australia, opened by the first test, is a development region scored with the frozen candidate models.

Population	Score	Region	Groups	k	Risk, k=1 (%)	Risk (%)	Used (%)	Unscreened (%)
mech	R7b	arso	12	1.24	6.7	5.8	94.7	8.2
mech	R7b	chmi	17	1.28	5.8	4.7	71.4	16.3
mech	R7b	gkd bayern	9	1.28	5.5	4.4	83.8	9.8
mech	R7b	lubw	7	1.28	5.4	4.4	81.6	11.7
mech	R7b	nve hydapi	17	1.26	6.3	5.0	71.1	16.2
mech	R7b	usgs huc02	11	1.24	6.5	5.5	71.4	16.3
mech	R7b	usgs huc03	15	1.28	5.2	4.3	86.6	8.4
mech	R7b	usgs huc04	9	1.28	3.9	3.2	85.5	6.4
mech	R7b	usgs huc05	11	1.28	5.6	4.3	68.7	14.8
mech	R7b	usgs huc07	5	1.28	6.0	5.2	70.2	15.5
mech	R7b	usgs huc10	7	1.28	5.6	4.8	69.2	14.1
mech	R7b	usgs huc17	18	1.28	4.3	3.4	76.6	11.1
mech	R7b	usgs huc18	5	1.24	6.5	5.7	78.1	13.1
mech	R7b	usgs huc19	8	1.28	5.1	4.1	70.5	11.1
mech	R7a	arso	12	1.45	8.2	6.3	96.0	8.2
mech	R7a	chmi	17	1.45	7.4	5.4	73.7	16.3
mech	R7a	gkd bayern	9	1.45	5.5	4.0	82.7	9.8
mech	R7a	lubw	7	1.45	5.9	4.2	80.8	11.7
mech	R7a	nve hydapi	17	1.45	7.3	5.1	72.8	16.2
mech	R7a	usgs huc02	11	1.45	7.2	5.7	70.7	16.3
mech	R7a	usgs huc03	15	1.45	5.4	4.1	85.8	8.4
mech	R7a	usgs huc04	9	1.45	4.4	3.4	86.5	6.4
mech	R7a	usgs huc05	11	1.45	5.9	4.2	68.8	14.8
mech	R7a	usgs huc07	5	1.45	7.2	5.9	70.5	15.5
mech	R7a	usgs huc10	7	1.45	5.9	4.6	67.9	14.1
mech	R7a	usgs huc17	18	1.45	5.0	3.8	78.1	11.1
mech	R7a	usgs huc18	5	1.45	6.7	4.8	76.1	13.1
mech	R7a	usgs huc19	8	1.45	7.2	5.9	73.6	11.1
deploy	R7b	arso	12	1.11	3.1	3.1	100.0	3.1
deploy	R7b	chmi	17	1.11	5.6	5.1	96.3	7.4
deploy	R7b	gkd bayern	9	1.12	1.0	1.0	100.0	1.0
deploy	R7b	lubw	7	1.12	1.5	1.5	100.0	1.5
deploy	R7b	nve hydapi	17	1.11	5.6	5.0	96.4	7.1
deploy	R7b	usgs huc02	11	1.12	2.8	2.8	100.0	2.8
deploy	R7b	usgs huc03	15	1.12	1.7	1.7	100.0	1.7
deploy	R7b	usgs huc04	9	1.12	0.6	0.6	100.0	0.6
deploy	R7b	usgs huc05	11	1.12	1.9	1.9	100.0	1.9
deploy	R7b	usgs huc07	5	1.11	3.0	3.0	100.0	3.0
deploy	R7b	usgs huc10	7	1.12	2.7	2.7	100.0	2.7
deploy	R7b	usgs huc17	18	1.12	2.1	2.1	100.0	2.1
deploy	R7b	usgs huc18	5	1.12	2.3	2.3	100.0	2.3
deploy	R7b	usgs huc19	8	1.12	2.3	2.3	100.0	2.3
deploy	R7a	arso	12	1.19	3.1	3.1	100.0	3.1
deploy	R7a	chmi	17	1.19	6.6	5.6	97.4	7.4
deploy	R7a	gkd bayern	9	1.30	1.0	1.0	100.0	1.0
deploy	R7a	lubw	7	1.30	1.5	1.5	100.0	1.5
deploy	R7a	nve hydapi	17	1.24	6.3	4.9	96.7	7.1
deploy	R7a	usgs huc02	11	1.30	2.8	2.8	100.0	2.8
deploy	R7a	usgs huc03	15	1.30	1.7	1.7	100.0	1.7
deploy	R7a	usgs huc04	9	1.30	0.6	0.6	100.0	0.6
deploy	R7a	usgs huc05	11	1.30	1.9	1.9	100.0	1.9
deploy	R7a	usgs huc07	5	1.19	3.0	3.0	100.0	3.0
deploy	R7a	usgs huc10	7	1.30	2.7	2.7	100.0	2.7
deploy	R7a	usgs huc17	18	1.30	2.1	2.1	100.0	2.1
deploy	R7a	usgs huc18	5	1.30	2.3	2.3	100.0	2.3
deploy	R7a	usgs huc19	8	1.30	2.3	2.3	100.0	2.3
deploy	R7b	Australia (opened, post hoc)	53	1.27	–	4.4	81.7	13.6
deploy	R7a	Australia (opened, post hoc)	53	1.45	–	4.0	80.5	13.6
mech	R7b	Australia (opened, post hoc)	53	1.27	–	4.9	62.6	19.5
mech	R7a	Australia (opened, post hoc)	53	1.45	–	4.5	61.6	19.5

Table S13. E8 downstream pipelines on 42 stations with at least 20 complete years (200 degradations each). Slopes in °C per decade relative to the complete record.

Scenario	Pipeline	Years	Trend bias	Trend RMSE	Cover (%)	Detect (%)	Mean H30 bias	Hot missed (%)
mcar	A0	24.4	0.0058	0.1691	–	65.3	0.0005	100.0
mcar	A1	29.7	0.0068	0.0524	–	66.6	−0.0046	18.8
mcar	A2	29.9	0.0031	0.0624	–	66.8	−0.0148	18.2
mcar	B	30.7	0.0004	0.0311	–	66.7	−0.0056	8.3
mcar	C	30.7	−0.0001	0.0183	96.1	66.7	−0.0009	5.0
mcar	C1	30.7	0.0001	0.0183	–	66.7	−0.0013	5.6
mnar1	A0	24.4	0.0108	0.1740	–	64.9	0.0413	100.0
mnar1	A1	29.7	0.0090	0.0565	–	66.7	−0.0013	22.0
mnar1	A2	30.0	0.0086	0.0649	–	66.7	−0.0128	21.2
mnar1	B	30.7	0.0022	0.0337	–	66.8	−0.0053	8.8
mnar1	C	30.7	−0.0000	0.0186	96.2	66.7	−0.0007	5.9
mnar1	C1	30.7	0.0004	0.0186	–	66.7	−0.0012	6.5
mnar2	A0	24.4	0.0084	0.1804	–	65.1	0.0308	100.0
mnar2	A1	29.7	0.0093	0.0569	–	66.7	−0.0020	21.7
mnar2	A2	29.9	0.0073	0.0647	–	66.7	−0.0141	21.5
mnar2	B	30.7	0.0021	0.0353	–	66.8	−0.0057	8.9
mnar2	C	30.7	0.0000	0.0194	95.9	66.7	−0.0005	6.1
mnar2	C1	30.7	0.0003	0.0198	–	66.7	−0.0010	6.8
realistic trend	A0	24.4	0.0077	0.1987	–	64.9	0.0502	100.0
realistic trend	A1	29.7	0.0088	0.0564	–	66.6	0.0007	20.5
realistic trend	A2	30.0	0.0087	0.0699	–	66.6	−0.0128	20.6
realistic trend	B	30.7	0.0018	0.0337	–	66.7	−0.0049	8.6
realistic trend	C	30.7	−0.0003	0.0184	96.1	66.7	−0.0002	5.8
realistic trend	C1	30.7	0.0000	0.0185	–	66.7	−0.0008	6.3

Table S14. E9 real incomplete station-years: share admitted by the label-free rule (R7b, applied within each region) and by four published rules (no truth is available; this is an application).

Provider	Years	Target-aware	USGS 311 d	WMO 11/5	Peak months	Two peak months
all	3876	99	54	46	89	61
arso	14	100	7	7	79	79
chmi	35	91	29	20	69	57
foen	1	100	100	0	100	100
gkd bayern	281	100	68	60	93	72
hubeau	36	72	0	0	50	56
lubw	526	100	54	44	93	52
nve hydapi	104	96	60	59	78	70
rws waterwebservices	8	100	62	38	88	75
syke	21	100	19	24	86	76
usgs	2850	99	54	46	89	61

Table S15. E11 score ablations for the plug-in failure, leave-one-region-out.

Population	Score	2.5%	5%	10%	AURC	Brier
deploy	geometry only	98.3	100.0	100.0	0.0019	0.0221
deploy	geometry plus layer	98.4	100.0	100.0	0.0018	0.0214
deploy	no simulation	99.0	100.0	100.0	0.0010	0.0157
deploy	full (R7b)	99.0	100.0	100.0	0.0009	0.0153
deploy	full plus old window summaries	99.0	100.0	100.0	0.0009	0.0152
deploy	full plus law	99.1	100.0	100.0	0.0009	0.0152
deploy	law only	98.9	100.0	100.0	0.0014	0.0180
deploy	replay law only	95.7	96.7	96.7	0.0013	0.0182
deploy	law v1 only	98.8	100.0	100.0	0.0014	0.0181
deploy	daily disagreement only	0.2	100.0	100.0	0.0353	–
deploy	target disagreement only	98.4	100.0	100.0	0.0024	–
mech	geometry only	39.5	49.4	71.6	0.0617	0.1227
mech	geometry plus layer	40.1	50.5	73.3	0.0578	0.1187
mech	no simulation	64.9	75.0	87.9	0.0313	0.0861
mech	full (R7b)	67.0	75.9	88.3	0.0301	0.0838
mech	full plus old window summaries	66.9	76.1	88.3	0.0300	0.0833
mech	full plus law	67.1	76.1	88.4	0.0300	0.0834
mech	law only	53.8	66.9	84.4	0.0413	0.1033
mech	replay law only	50.0	63.1	81.9	0.0417	0.1052
mech	law v1 only	53.4	66.6	84.3	0.0415	0.1033
mech	daily disagreement only	0.3	1.3	8.9	0.1333	–
mech	target disagreement only	47.6	57.7	75.9	0.0525	–

Table S16. Sensitivity to the tolerance (0.25, 0.5 and 1.0 °C): unscreened failure rates and retention at 2.5% and 5% realised risk of the target-aware judgment (D2* value) and of the best fixed published rule (recorded-day or plug-in value).

Tolerance	Population	Fail P0	Fail recorded	Fail D2*	Target 2.5%	Rule 2.5%	Target 5%	Rule 5%
0.25	deploy	6.1	8.9	4.7	96.9	90.5	100.0	90.5
0.25	mech	29.5	46.4	25.4	52.4	28.6	60.0	28.6
0.50	deploy	3.3	7.0	2.4	100.0	90.5	100.0	90.5
0.50	mech	17.0	38.7	12.0	69.9	28.6	80.9	28.6
1.00	deploy	1.5	5.5	0.9	100.0	90.5	100.0	90.5
1.00	mech	6.5	28.0	3.3	97.9	43.8	100.0	47.6

Table S17. Post hoc conditional calibration of the target-aware score (D2* failure probability) under real gap calendars after the first test: observed failure rate by score bin in the source (frozen models, in-sample) and in Australia.

Domain	Score bin	n	Mean score	Observed failure (%)
source	(−0.01, 0.0]	28,703	0.000	0.0
source	(0.05, 0.1]	462	0.074	6.5
source	(0.1, 0.2]	481	0.152	17.0
source	(0.2, 0.3]	428	0.248	25.2
source	(0.3, 0.5]	657	0.395	43.4
source	(0.5, 1.0]	447	0.594	65.8
australia	(−0.01, 0.0]	5,085	0.000	0.7
australia	(0.05, 0.1]	503	0.074	10.5
australia	(0.1, 0.2]	548	0.154	19.5
australia	(0.2, 0.3]	563	0.251	33.6
australia	(0.3, 0.5]	891	0.395	47.9
australia	(0.5, 1.0]	560	0.605	68.8

Table S18. Corrected hemisphere and season definitions (M1). Source: round-1 and round-2 cases joined case by case. Australia: the round-1 test cases against the round-2 development cases (post hoc).

Quantity	Value	Of / law
cases joined	145,684	145,684
identical masks	1	
identical true H30	1	
identical plug-in H30	1	
identical layer	1	
stations whose warm season is not May-September	163	581
cases with a changed warm-season observed fraction	0.118	
mean absolute change of warm-season observed fraction	0.015	
median relative change of warm-season residual sd	0.000	
stations whose degree-day months are not June-September	40	581
Australia, round 1 (calendar year, May-Sep), calendar: window at year edge / plug-in failure / mean error / law v2, v2p	30.8 / 16.7 / −0.025	
Australia, round 1 (calendar year, May-Sep), crossed: window at year edge / plug-in failure / mean error / law v2, v2p	30.8 / 22.8 / −0.008	
Australia, round 2 (July-June hot year, station seasons), calendar: window at year edge / plug-in failure / mean error / law v2, v2p	1.1 / 15.8 / −0.024	15.3 / 16.2
Australia, round 2 (July-June hot year, station seasons), crossed: window at year edge / plug-in failure / mean error / law v2, v2p	1.1 / 24.8 / −0.000	22.3 / 23.9

Table S19. Split-conformal calibration of the D2* quantiles by decision-time difficulty group (share of the climatological peak window hidden: bin 0 = none, bins 1–4 = quarters; information layer), leave-one-region-out. Coverage and width of the 90% interval, all source cases.

Group	Cases	Groups	Cover raw (%)	Cover calibrated (%)	Width raw	Width calibrated
bin0, air ridge	10,526	176	98.6	97.2	0.07	0.07
bin0, donor ridge	50,515	166	99.0	97.4	0.04	0.04
bin0, other	377	16	98.9	95.0	0.10	0.10
bin1, air ridge	1,944	156	91.4	91.3	0.55	0.55
bin1, donor ridge	7,161	165	91.6	91.3	0.36	0.36
bin2, air ridge	3,689	168	88.6	89.9	0.88	0.93
bin2, donor ridge	11,669	166	87.7	89.9	0.64	0.68
bin3, air ridge	3,138	156	87.9	90.0	1.15	1.22
bin3, donor ridge	10,023	165	86.5	89.9	0.83	0.90
bin4, air ridge	10,821	166	85.4	90.1	1.60	1.83
bin4, donor ridge	33,591	164	84.3	89.8	1.10	1.26
bin4, other	317	16	79.2	86.4	2.48	3.34

Table S20. Second test on new USGS stations (second preregistration, N1), recomputed after the change of random seeds (Text S6). Admission by the label-free rule over all test cases of each population; R7b is primary, R7a and k=1 are secondary. Upper bounds resample independence groups; strata are fixed from availability metadata.

Population	Score	Factor	Cases	Groups	Used (%)	Risk (%)	Upper (%)	Unscreened (%)	High missing (%)	Low missing (%)
deploy	R7b	k	2648	40	99.6	4.5	6.4	4.8	6.2	3.6
deploy	R7b	k1	2648	40	100.0	4.8	6.8	4.8	6.3	3.9
deploy	R7a	k	2648	40	98.9	4.3	6.1	4.8	6.1	3.3
deploy	R7a	k1	2648	40	100.0	4.8	6.8	4.8	6.3	3.9
mech	R7b	k	9297	40	74.6	5.6	6.8	14.9	6.4	5.2
mech	R7b	k1	9297	40	78.7	7.0	8.5	14.9	7.5	6.7
mech	R7a	k	9297	40	75.2	5.7	6.9	14.9	6.3	5.4
mech	R7a	k1	9297	40	81.1	7.6	9.1	14.9	8.4	7.2

Table S21. Where the downstream gain comes from (M5). Upper rows: station H30 trend RMSE (°C per decade) for the 90% rule (A1), the plug-in with all years (B) and multiple imputation (C), the share of the A1-to-C gain obtained by keeping years (A1 to B), and station-level RMSE ratios (median and quartiles). Lower rows: regional mean H30 trend of the same 19 Baden-Württemberg stations with a 95% interval over independence groups (real incomplete years, no truth).

Scenario or region	A1	B	C	From keeping years	B/A1	C/B
mcar	0.0524	0.0311	0.0183	63	0.46 [0.23, 0.71]	0.76 [0.60, 0.92]
mnar1	0.0565	0.0337	0.0186	60	0.40 [0.18, 0.53]	0.81 [0.66, 0.89]
mnar2	0.0569	0.0353	0.0194	58	0.39 [0.20, 0.61]	0.76 [0.61, 0.87]
realistic trend	0.0564	0.0337	0.0184	60	0.46 [0.19, 0.58]	0.80 [0.67, 0.87]
Baden-Württemberg, all years, A0	0.72	[0.50, 1.05]				
Baden-Württemberg, all years, A1	0.35	[0.23, 0.72]				
Baden-Württemberg, all years, B	0.33	[0.23, 0.62]				
Baden-Württemberg, all years, C	0.32	[0.22, 0.62]				
Baden-Württemberg, without 2003, A0	0.72	[0.50, 1.05]				
Baden-Württemberg, without 2003, A1	0.38	[0.26, 0.75]				
Baden-Württemberg, without 2003, B	0.36	[0.26, 0.64]				
Baden-Württemberg, without 2003, C	0.35	[0.25, 0.64]				

Table S22. S1: annual 7-day minimum flow on 255 perennial USGS streams (climatic years April–March). Bias is the geometric-mean relative error, $\exp(\overline{\log \hat{Q} - \log Q}) - 1$; failure means a log error beyond 18%. Trend RMSE of log 7Q per decade: skip incomplete years 0.0217, plug-in 0.0099, multiple imputation 0.0087.

Population	Stratum	Estimator	n	Bias (%)	MAE (log)	Failure (%)	Cover90 (%)
mech	all	P0 plug-in	80,931	1.0	0.056	10.9	–
mech	all	recorded days	80,931	6.7	0.065	12.7	–
mech	all	D2 target-aware	80,931	2.5	0.050	9.3	93.2
mech	low-flow window at least half hidden	P0 plug-in	41,132	1.7	0.083	16.0	–
mech	low-flow window at least half hidden	recorded days	41,132	10.6	0.100	19.8	–
mech	low-flow window at least half hidden	D2 target-aware	41,132	3.6	0.074	13.9	91.0
mech	half hidden, neighbour layer	P0 plug-in	33,609	1.0	0.084	16.4	–
mech	half hidden, neighbour layer	recorded days	33,609	10.9	0.104	20.5	–
mech	half hidden, neighbour layer	D2 target-aware	33,609	3.3	0.075	13.9	90.5
mech	half hidden, interpolation layer	P0 plug-in	7,523	4.7	0.077	14.4	–
mech	half hidden, interpolation layer	recorded days	7,523	8.9	0.085	16.5	–
mech	half hidden, interpolation layer	D2 target-aware	7,523	4.8	0.073	13.9	93.4
deploy	all	P0 plug-in	53,994	2.8	0.075	13.8	–
deploy	all	recorded days	53,994	15.0	0.140	20.6	–
deploy	all	D2 target-aware	53,994	2.9	0.068	12.5	93.3
deploy	low-flow window at least half hidden	P0 plug-in	17,707	7.5	0.172	31.5	–
deploy	low-flow window at least half hidden	recorded days	17,707	43.9	0.364	51.2	–
deploy	low-flow window at least half hidden	D2 target-aware	17,707	6.0	0.156	29.0	88.4
deploy	half hidden, neighbour layer	P0 plug-in	14,394	4.3	0.142	28.1	–
deploy	half hidden, neighbour layer	recorded days	14,394	41.9	0.350	50.8	–
deploy	half hidden, neighbour layer	D2 target-aware	14,394	4.5	0.134	25.8	88.3
deploy	half hidden, interpolation layer	P0 plug-in	3,313	22.5	0.302	46.2	–
deploy	half hidden, interpolation layer	recorded days	3,313	52.8	0.424	52.8	–
deploy	half hidden, interpolation layer	D2 target-aware	3,313	12.5	0.253	43.2	88.7